Aditya Mate, PhD

Short C.V.

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Link to full CV

Summary

Applied Scientist at **Microsoft** and **Harvard PhD** (computer science) with **8 years of AI/ML research** experience. Research focuses on LLMs, reasoning agents, evaluation, safety, routing and decision-making under uncertainty. Prior research experience at **Google Research** and **IBM Research**. Published at NeurIPS, ICML, AAAI, AAMAS, EMNLP etc. with experience translating research into deployed ML systems in product settings.

Education

2018–2023 **Harvard University**, Cambridge, MA
Ph.D. in Computer Science | Thesis: Actualizing Impact of AI and Optimization in Public Health.

2013–2018 **Indian Institute of Technology Bombay**, Mumbai
B.Tech & M.Tech in Electrical Engineering | Minor: Computer Science and Engineering.

Industry Experience

July'23–Present **Microsoft**, Cambridge, MA.
Applied Scientist 2 | Manager: Dr. Yingnong Dang

- Currently work in the AI Model & Solution team within Azure, developing probabilistic models and reasoning agents to improve the quality and reliability of Azure services.
- Selected at Microsoft's Office of the CTO through their MAIDAP Rotation Program, collaborating with four product organizations across Strategic Missions and Technologies, Microsoft 365, GenAI, and Microsoft Search, Assistant and Intelligence (MSAI) to infuse AI/ML research in product problems.
- Conducted research and developed ML/LLM solutions across LLM evaluation, fine-tuning and safety, cost-aware LLM routing, RAG/Graph RAG, and reasoning over time-series data, translating research into deployed ML systems in product settings.

2021–2022 **Google Research**, United States / India.
Research Intern / Student Researcher | Managers: Philip Nelson, Dr. Aparna Taneja

- Led the development of a novel approach for evaluating algorithmic resource-allocation policies in randomized trials, enabling more accurate causal estimates; published at **ICML 2023**.
- Led the field study and deployment of a restless multi-armed bandit system for improving maternal and child health outcomes, translating ML research into a real-world intervention program; published at **AAAI 2022** and recognized with an **IAAI 2023 Innovative Application Award**.
- Contributed to parallel research efforts in **decision-focused learning** and **sequential decision-making under uncertainty** and mentored related research projects, resulting in publications at **AAMAS 2023**, **AAAI 2023**, and EAAMO 2022.

June–Sept'21 **IBM Research**, Yorktown Heights, NY (remote).
Research Intern | Manager: Dr. Kush R. Varshney

- Data-driven methods for planning and resource allocation for social-impact non-profits **(paper)**.

Selected Academic Achievements

2023 IAAI 'Innovative Application' Award for deployed AI system in maternal healthcare.

2022 INFORMS "Doing Good with Good OR" Competition Runner-up prize.

2021 **Best Paper Award** at NeurIPS'21 Workshop on Machine Learning in Public Health (MLPH).

2020 **Best Lightning Paper Award** at NeurIPS'20 Workshop on Machine Learning in Public Health.

2020 **Best Paper Award** (theme) at NeurIPS'20 Workshop on Machine Learning for Health (ML4H).

Selected Publications

Generative AI / LLM / Modern AI Systems

CAIS 2026 — Emmanuel Boateng, et al. **Aditya Mate**, Ehi Nosakhare.
"SAPO: Secure Automated Prompt Optimization via Multi-Agent Collaboration", *ACM Conference on AI and Agentic Systems (CAIS) 2026*, [paper].

EMNLP 2025 — Anh Pham et al., **Aditya Mate**, Ehimwenma Nosakhare, Soundararajan Srinivasan
"How to Fine-Tune Safely on a Budget: Model Adaptation Using Minimal Resources", *Empirical Methods in Natural Language Processing (EMNLP) 2025 Industry Track*, [paper].

MSJAR 2025 — Tian Xia, **Aditya Mate***, Regina Ceballos*, et al.
"Building Gold-Standard Datasets for AI Quality Evaluation", *Microsoft Journal on Applied Research (MSJAR) volume 22*, *Equal contribution, [MSFT confidential].

NeurIPS 2025 workshop — Roshini Pulishetty* et al., **Aditya Mate**, Somya Chatterjee et al.
"One Head, Many Models: Cross-Attention Routing for Cost-Aware LLM Selection", *Workshop on Evaluating the Evolving LLM Lifecycle NeurIPS 2025*, **Equal contribution*.

MSJAR 2023 — Newman Cheng et al., **Aditya Mate**, Jonathan Larson
"Graph Based LLM Agents for Complex Reasoning Tasks", *Microsoft Journal of Applied Research (MSJAR) Volume 20, (2023)*, [paper internal to Microsoft].

Machine Learning / Reinforcement Learning / AI Decision-Making (All First Author)

ICML 2023 — **Aditya Mate**, Bryan Wilder, Aparna Taneja and Milind Tambe
"Improved Policy Evaluation for Randomized Trials of Algorithmic Resource Allocation", *International Conference on Machine Learning (ICML) 2023*, [paper].

Book Chapter 2023 — **Aditya Mate**, Shresth Verma, Gargi Singh, Aparna Taneja and Milind Tambe
"Field Study in Deploying Restless Multi-Armed Bandits: Assisting Non-Profits in Improving Maternal and Child Health", [link].

AAAI 2022 — **Aditya Mate***, Lovish Madaan*, Aparna Taneja, et al.
"Field Study and Deployment of Restless Multi-Armed Bandits for Improving Maternal and Child Health Outcomes", *AAAI Conference on Artificial Intelligence 2022, *Equal contribution*, [paper],
• *Also appeared at **NeurIPS 2021 workshop** on Machine Learning in Public Health (MLPH).*
Won **Best Paper Award**

AAMAS 2022 — **Aditya Mate**, Arpita Biswas, Christoph Siebenbrunner, Susobhan Ghosh and Milind Tambe.
"Efficient Algorithms for Finite Horizon and Streaming Restless Multi-Armed Bandit Problems", *International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2022*, [paper].

AAMAS 2021 — **Aditya Mate**, Andrew Perrault and Milind Tambe.
"Risk-Sensitive Interventions in Public Health: Planning with Restless Multi-Armed Bandits", *International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2021*, [paper].

NeurIPS 2020 — **Aditya Mate***, Jackson Killian*, Haifeng Xu, Andrew Perrault and Milind Tambe.
"Collapsing Bandits and Their Application to Public Health Interventions", *Advances in Neural and Information Processing Systems (NeurIPS) 2020 *Equal contribution*, [paper].

Professional Service

- **Area Chair / Senior PC** for ICLR 2027, AAAI 2027, ICML (2026, 2025), NeurIPS 2026, IJCAI 2026, AAMAS 2025. Reviewer for several top-tier AI research conferences in prior years.

Press Coverage & Research Impact

- Research covered in top-tier news outlets and AI blogs including: Business Insider, Google AI Blog, Medical Xpress, Nature Research Asia newsletter, Harvard SEAS newsletter, Sakal Media